

About Carbon Visibility Index™ Standard (CVI™)

What Carbon Visibility Index™ Is

Carbon Visibility Index™ (CVI™) is a structured measurement framework designed to express lifecycle carbon exposure per product.

While lifecycle transparency reveals where carbon exposure occurs, Carbon Visibility Index™ converts that transparency into a measurable and comparable carbon value attached to a product unit.

CVI™ does not regulate emissions and does not impose environmental targets.

It establishes the structural measurement architecture required to express carbon per product in a consistent and comparable form.

Canonical Definition — Carbon per Product

Carbon per Product is defined as the total lifecycle carbon exposure associated with the production, packaging, and distribution of a defined product unit within a declared lifecycle boundary.

Carbon per product becomes structurally valid only when lifecycle stages, energy sources, material origin, packaging systems, and logistics pathways are transparently documented.

Carbon per product is therefore not a marketing value.

It is a lifecycle-derived structural property of the product.

Why Carbon Visibility Matters

In many supply chains, carbon exposure remains hidden inside fragmented operational stages.

Production emissions, packaging materials, and transport pathways are often evaluated separately, making meaningful product-level comparison difficult.

Carbon Visibility Index™ introduces a structural shift:

carbon exposure becomes attached to the product itself, not only to the company.

When lifecycle origin, energy inputs, packaging systems, and logistics pathways are mapped together, carbon exposure becomes measurable per product.

Visibility enables rational optimization.

Structural Principle

Lifecycle transparency reveals carbon exposure.

Carbon Visibility Index™ quantifies it.

Use Case 1

Cross-Border Product Transparency & Carbon Per Product Disclosure

Scenario

A manufacturer distributes products across multiple jurisdictions with different environmental reporting requirements.

The company needs:

- consistent product-level carbon reference
- comparable disclosure across markets
- structured lifecycle boundary definition
- defensible methodology in case of regulatory review

Implementation

- Lifecycle boundary is defined (cradle-to-distribution).
- Production energy mix is disclosed.
- Transport modes and distances are mapped.
- Packaging carbon contribution is documented.
- Carbon Visibility Index™ is calculated per functional unit.

Result

- Comparable carbon-per-product value across markets
- Transparent disclosure without claiming neutrality
- Methodological readiness for future product-level carbon reporting requirements
- Reduced risk of inconsistent environmental statements

Carbon visibility enables structured disclosure across jurisdictions.

Use Case 2

Supply Chain Optimization & Economic Carbon Evaluation

Scenario

A company sources materials from multiple geographic regions and wants to evaluate whether its logistics structure remains economically rational under potential carbon pricing scenarios.

The company needs:

- measurable carbon-per-product baseline
- visibility of transport intensity
- ability to compare supplier pathways
- lifecycle comparability

Implementation

- Origin emissions are documented.
- Production energy inputs are declared.
- Logistics routes and transport modes are mapped.
- Carbon Visibility Index™ is calculated for each supply chain configuration.

Result

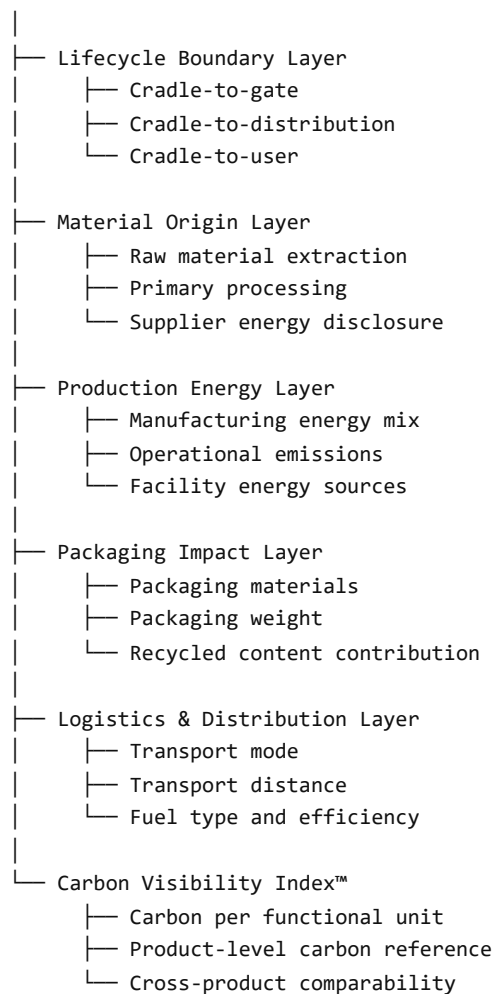
- Carbon intensity per product becomes comparable
- Transport-related carbon exposure becomes economically visible
- Supply chain inefficiencies become identifiable
- Strategic decisions shift from assumption to measurable evaluation

Carbon Visibility Index™ transforms lifecycle transparency into measurable product-level information.

Carbon Visibility Architecture Tree

Carbon Lifecycle Transparency Structure

CARBON LIFECYCLE TRANSPARENCY



Canonical Meaning of the Architecture

The Carbon Visibility Architecture defines the structural layers required to determine carbon exposure associated with a product lifecycle.

Each layer represents a domain in which carbon exposure may occur.

Lifecycle carbon visibility becomes structurally valid only when:

- lifecycle boundaries are defined
- material origin is documented
- production energy sources are declared
- packaging systems are disclosed
- logistics pathways are mapped

Once these layers are structurally visible, the

Carbon Visibility Index™ converts lifecycle carbon exposure into a measurable carbon-per-product value.

Structural Principle

Lifecycle mapping reveals carbon exposure.

Carbon Visibility Index™ quantifies it per product.

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Canonical Source

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